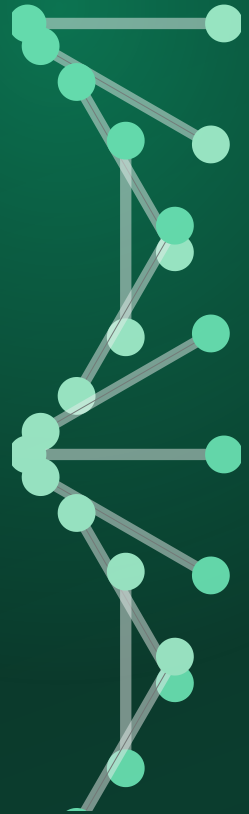




Deccan Education Society's Fergusson College
Department of Biology



STUDENT & PARENT HANDBOOK · ADMISSIONS 2026

Why Choose Biology?

Your future-proof choice for 2026 — and the next 100 years.

"The science of living, breathing, and thriving."

Std. XI / FYJC Science

Welcome

Biology is the technology of the next 100 years.

While machines automate the routine, the living world — genes, cells, brains and ecosystems — becomes humanity's biggest frontier. Choose Biology and build a career that AI can't replace and the future can't do without.

10M+

Healthcare jobs needed by 2030 (WHO)

\$4T+

Global bio-economy, projected

200M+

Protein structures mapped by AI

2020

Nobel Prize for CRISPR

Inside this booklet

1. 10 Reasons to Choose Biology
2. Breakthroughs Biology Students Build
3. How Future-Proof Is Your Subject?
4. One Subject. 50+ Careers.
5. A Message for Parents
6. Meet Your Faculty
7. Where Biology Can Take You

10 Reasons to Choose Biology



01

Careers AI simply can't replace

Doctors, surgeons, nurses and caregivers rely on human judgement, empathy and a steady hand — the exact things machines can't reproduce. A robot can assist; it can't reassure a frightened patient.

Care & surgery rank among the hardest jobs to automate



02

Rising healthcare demand = job security

The world is ageing and getting more health-conscious. More people, living longer, means a permanent, recession-proof need for medical and health professionals everywhere.

1 in 6 people will be 60+ by 2030 (WHO)



03

Biotech & genetic engineering — the next trillion-dollar industries

CRISPR gene editing, gene therapy and personalised medicine are turning science fiction into clinics. This is where the biggest companies and breakthroughs of your lifetime will be built — and they need biology-trained minds.

Biotech is among the fastest-growing global sectors



04

The most respected — and best paid — careers

Doctor, surgeon, dentist, pharmacist, researcher: biology opens the doors to professions that society trusts most and rewards generously, for a whole working lifetime.

Healthcare professions top respect & income surveys



05

Biology + AI = the real future

AI doesn't replace biology — it supercharges it. Bioinformatics, computational biology and AI drug discovery are exploding fields where biology students who understand data become unstoppable.

AI + biology is one of the hottest skill combinations on earth



06

Opportunities that travel the world

A biology degree is respected and in demand globally. Doctors, researchers and biotech specialists are welcomed in the US, UK, Germany, Canada and the Gulf — your career needn't stop at the border.

Skilled-health migration pathways in 5+ regions



07

Solve humanity's biggest problems

Pandemics, cancer, climate change, feeding 8 billion people — every one of these is a biology problem. Choosing biology means choosing work that genuinely matters for the planet.

The defining challenges of this century are biological



08

One subject, dozens of doorways

Medicine, agriculture, environment, forensics, sports science, veterinary science and more — biology is the foundation that keeps your options wide open long after Class 11.

50+ career fields branch from one biology base



09

Build the next great company

Biotech start-ups, health-tech apps, medical devices and pharma are magnets for founders and investors. Biology gives entrepreneurs the one thing money can't buy — deep understanding of life itself.

Health & bio start-ups attract record global funding



10

The science of life itself — never obsolete

Technologies come and go, but understanding our body, disease and the living world is timeless. Biology is knowledge that will matter as much in 2125 as it does today.

Life science is the one field that can't go out of date

Breakthroughs Biology Students Build

A • Human Body & Medicine Rewriting what it means to heal



Human Genome Project

Mapped all 3 billion letters of human DNA — the instruction manual for a human being.



CRISPR gene editing

Molecular "scissors" that rewrite faulty DNA to cure genetic disease — Nobel Prize 2020.



mRNA vaccines (COVID-19)

The fastest vaccines ever made — estimated to have saved roughly 20 million lives in their first year.



Stem cells & lab-grown organs

Growing replacement tissue, skin and even mini-organs to repair the human body.



Bionic limbs & brain-computer interfaces

Neuralink and others are connecting minds directly to machines and giving movement back to the paralysed.



Personalised medicine & immunotherapy

Treatment tuned to your own DNA, and immune cells trained to hunt and kill cancer.



IVF & fertility science

Lab-assisted conception — the Nobel-winning IVF technique — has helped millions of families have a baby.



3D bioprinting

Printing living cells layer by layer to build tissue patches and, one day, whole organs.

B • Botany & Nature Engineering a greener, fed and thriving planet



Climate-resilient & GM crops

Crops engineered to survive drought, pests and heat — feeding billions on a warming planet.



Vertical farming & cultivated meat

Food grown in stacked indoor towers — and real meat grown from cells, not animals.



Biofuels & algae energy

Living organisms like algae turned into clean, renewable fuel for a low-carbon future.



Conservation biology

Saving endangered species and whole ecosystems from disappearing forever.



Synthetic biology

Engineering living organisms like programmable machines to make medicines, materials and more.



Seed banks & gene vaults

Safeguarding the planet's crop and plant DNA so nothing is lost to future generations.



Mycology & fungi materials

Growing leather, packaging, food and medicine from mushrooms and fungi.



Coral reef restoration

Rebuilding dying reefs that shelter a quarter of all ocean life and protect coastlines.

C · Biology in Space Life sciences are leaving the planet



Plants on the ISS & Mars

Astronauts already grow fresh lettuce in orbit — the first step to gardens on Mars.



Astrobiology

The search for life beyond Earth — on Mars, ice moons and distant planets.



The body in zero gravity

Studying how muscle, bone, blood and the heart change when astronauts live in space.



Space agriculture

Growing enough food to keep crews alive on missions that last years.



Space medicine & radiation biology

Shielding astronauts from cosmic radiation and keeping their bodies healthy on long missions.



Algae & microbe life-support

Living systems that recycle air, water and waste so crews can breathe and drink in space.

D · Biology + AI & Future Tech Where data science meets the living world



AlphaFold

AI cracked a 50-year mystery — predicting the shape of proteins, and mapped 200 million+ of them.



AI-driven drug discovery

AI designs and tests new medicines in months instead of years, slashing the cost of cures.



Bioinformatics & big-data biology

Turning oceans of genetic and medical data into real discoveries.



Wearable & digital health

Your watch becomes a 24/7 doctor — tracking heart, sleep and oxygen to catch illness early.

How Future-Proof Is Your Subject?



Biology



Geography



IT



Computer Sci.

AI can't replace the human touch



Lifelong job security



Hands-on, real-world work



Solves humanity's big problems



Works with AI as a tool



Global career mobility



Recession-proof demand



Emotional & human connection



Entrepreneurship potential



Knowledge that never goes obsolete



One lens, not a verdict. These scores illustrate resilience to automation over the coming decades — not the worth of any subject or the talent of its teachers. Geography, IT and Computer Science remain outstanding, in-demand choices, and the best path is the one a student truly loves. Biology simply happens to sit furthest from what AI can fully replace.

One Subject. 50+ Careers.



Doctor (MBBS)

Diagnose and treat patients



Surgeon

Perform life-saving operations



Dentist (BDS)

Teeth, gums & oral health



Pharmacist

Medicines and drug safety



Physiotherapist

Movement and rehabilitation



Nurse

Frontline patient care



Veterinarian

Animal health and surgery



Biotechnologist

Engineer living systems



Geneticist

Genes, heredity and DNA



Microbiologist

Microbes, bacteria, viruses



Biochemist

The chemistry of life



Molecular Biologist

DNA, RNA and proteins



Immunologist

Immune system and vaccines



Bioinformatician

Biology meets big data



Pharmacologist

How drugs act on the body



Clinical Researcher

Trials for new cures



Epidemiologist

Track and stop outbreaks



Pathologist

Diagnose disease from samples



Radiologist

Medical imaging and scans



Nutritionist / Dietician

Food and diet science



Agricultural Scientist

Better crops and farming



Horticulturist

Plants, orchards and gardens



Botanist

The science of plant life



Zoologist

The science of animal life



Marine Biologist

Ocean life and reefs



Wildlife Biologist

Conservation of species



Environmental Scientist

Protecting ecosystems



Ecologist

The balance of nature

**Forensic Scientist**

Solve crimes with biology

**Food Technologist**

Safer, better food

**Genetic Counselor**

Guide on inherited risk

**Biomedical Engineer**

Design medical devices

**Bioprocess Engineer**

Scale up bio-manufacturing

**Pharma Production**

Make medicines at scale

**Public Health Officer**

Community health programs

**Hospital Administrator**

Run healthcare systems

**Medical Writer**

Communicate science clearly

**Teacher / Professor**

Educate the next generation

**Research Scientist**

Push the frontiers

**Stem Cell Scientist**

Regenerative medicine

**Cancer Researcher**

Oncology and new therapies

**Neuroscientist**

Brain and nervous system

**Synthetic Biologist**

Design new biology

**Astrobiologist**

Search for life in space

**Sports Scientist**

Human performance

**Optometrist**

Eye care and vision

**Audiologist**

Hearing and balance

**Toxicologist**

Poisons and safety science

**Biotech Entrepreneur**

Build bio start-ups

**Drug Discovery Scientist**

Invent new medicines

A Message for Parents

Will this career still be safe in an age of AI?

It's the right question to ask in 2026 — and biology has the most reassuring answer of any subject your child can choose.

"Won't AI and robots take these jobs?"

AI automates tasks — it doesn't replace human care, judgement and discovery . A machine can scan a report, but it can't comfort a patient, perform delicate surgery, run a wet-lab experiment, or decide which life to save first.

In biology, AI becomes your child's tool, not their replacement . The people who combine biology with these new tools won't be automated away — they'll be the ones in highest demand.

Permanent job security

People will always fall ill, age and need food. The demand for health and life-science skills never disappears.

Strong, growing earnings

Medicine, biotech, pharma and research are among the best-paid, most stable careers — in India and abroad.

Many doors, one decision

From doctor to data-biologist to entrepreneur — biology keeps options open instead of narrowing them.

Respect & purpose

Few careers earn the trust and gratitude of society like saving lives and protecting our planet.

Meet Your Faculty

Our teachers don't just cover the syllabus — they teach the science of life with real passion, and every one of them stays up to date with the latest AI tools. With us you'll learn to use AI to power your Biology career, not compete with it. Friendly mentors, advanced learning, your growth first.



Prof. Vijay Zambare



Prof. Namdev Andhale



Prof. Pooja Nashikkar



Prof. Shweta Laddha



Prof. Smita Patil



Prof. Vaibhavi Patil



Prof. Tarini Joshi



Prof. Prachi Kambale



Prof. Gauri Gaikwad



Prof. Amruta Waghmare

Where Biology Can Take You

Syngene

Emcure Pharmaceuticals

Asian Paints

KPIT

D.E. Shaw

Harbinger

Josh Software

AlgoAnalytics

Ellicium

WisdmLabs

Tech Prescient

RippleHire

SpringCT

Omniscient

NAAC A+ Grade · UGC Special Heritage Status · Autonomous



Choose Biology

Take the first step today. Talk to our Biology faculty and secure your place for the 2026 academic year.

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Autonomous